TFOW: Detecting Anomalous Order Flow in Crypto Limit Order Books

Anonymous submission

Abstract

The limit order book (LOB) is the core mechanism of crypto centralized exchanges by continuously recording bids and asks to facilitate price discovery and maintain liquidity, yet it is also the primary channel through which most forms of cryptocurrency market manipulation are executed. Prior studies have examined pump-and-dump schemes and wash tradings which undermine the integrity of the market, but these approaches operate at the token or exchange level, leaving open the critical question of which specific orders or order flows are manipulative. Without this granularity, regulators and exchanges lack actionable countermeasures against specific manipulators. To address this gap, we introduce the TFOW framework, which constructs atomic order events from synchronized trade and order book update data, and models their dynamics with a set-valued marked temporal point process (set-valued marked temporal point process (set-mtpp)). Leveraging the learned intensity function, we rescale inter-arrival times and apply the sum-of-squared-spacings (3S) statistic for principled anomaly detection at the sequence level. Detected anomalies are then localized through set-mtpp-based probabilistic queries, which provide timestamp-level resolution and surface rare temporal motifs characteristic of manipulative behavior.

Our TFOW [PLACEHOLDER FOR RESULTS].

Overall, TFOW delivers interpretable order-level anomaly detection and equips regulators and exchanges with actionable tools for market integrity, offering a generalizable framework for identifying anomalous event flows in continuous time.

Intention, for market making, it is essential to have a stable simulation model. If we can predict well, we can also use the model in prediction to better capture the fill probability and adversary events 2. s2p2 has achieved state of art in per event total log-likelihood, RMSE of the next event time prediction, and next mark classificcation accuracy. But no stability,

backgroud

Temporal Point Processes and Order Book Modeling

Temporal point process (TPP) model event arrivals in continuous time via a conditional intensity $\lambda^*(t)$ (Shchur et al. 2021b). Marked TPPs (MTPPs) (Upadhyay, De, and Gomez Rodriguez 2018) enrich this framework by attaching categorical marks to each event, capturing both timing and type. In financial markets, MTPPs are widely applied to model limit order book dynamics, where marks naturally encode order types (e.g., limit, cancel, market). A prominent example is the Hawkes process, a self-exciting MTPP extensively used for order book modeling due to its ability to capture clustering in order arrivals (Jain et al. 2024a, 2025). The neural Hawkes process (Mei and Eisner 2017) extends this idea with neural architectures that learn nonlinear, history-dependent intensities, outperforming classical parametric forms. However, these models assume a single event per timestamp, an unrealistic constraint in limit order books where multiple actions often co-occur. To overcome this, Chang et al. (Chang, Boyd, and Smyth 2024b) introduce the set-valued MTPP (Set-MTPP), which generalizes marks from singletons to sets, enabling principled modeling of simultaneous market events. This foundation is critical for capturing realistic order flow and underpins our anomaly detection framework.

Market Participants and Manipulators

In centralized exchanges and under normal trading conditions, order flows predominantly arise from two types of participants: investors, who submit directional orders to express market views, and market makers, who continuously quote on both sides of the book to provide liquidity while managing inventory risk (Ganesh et al.). Under this natural market composition, the aggregate order flow exhibits a relatively stable probabilistic structure. However, when **manipulators** intervene—through practices such as pump-and-dump schemes or wash trading—they deliberately inject strategic orders into this equilibrium, distorting the underlying dynamics generated by investors and market makers and undermining the integrity of the market.

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Anomaly Detection

1. deviation from the model. First order moment, second order moment 2. Querying

Method

Stationary-Mean Calibration for Set-Valued Hawkes Ground Intensity

We model order-book event streams as a sequence of set-valued continuous-time events

$$\mathcal{H}_T = \{(t_i, x_i)\}_{i=1}^{N_T}, \qquad x_i \in \mathcal{X} := \mathcal{P}(\{1, \ldots, K\}),$$

where x_i denotes the subset of event types that occur simultaneously at time t_i. Following the set-valued temporal point process formulation of Chang, Boyd, and Smyth (Chang, Boyd, and Smyth 2024a), we separate the temporal arrival process from the conditional set distribution. Specifically, the set-specific intensity is written as

$$\lambda_x^*(t) = \lambda^*(t)\, p^*(x \mid t), \qquad x \in \mathcal{X},$$

where $\lambda^*(t)$ is the ground intensity of set-event arrivals and $p^*(x \mid t)$ is a normalized distribution over subsets. Thus,

$$\sum_{x \in \mathcal{X}} \lambda_x^*(t) = \lambda^*(t) \sum_{x \in \mathcal{X}} p^*(x \mid t) = \lambda^*(t).$$

The set head is therefore rate-neutral: it redistributes probability mass across possible simultaneous event sets but does not change the total arrival rate.

In the Dynamic Bernoulli version of Chang et al.'s set model, the conditional distribution over subsets is factorized as

$$p_\omega(x \mid t, h_\theta(t)) = \prod_{k=1}^{K} \rho_k(t)^{\mathbf{1}(k \in x)} \big(1 - \rho_k(t)\big)^{\mathbf{1}(k \notin x)},$$

where

$$\rho_k(t) = \sigma\big(v_k^\top n(h_\theta(t)) + b_k\big)\,.$$

Here, $h_\theta(t)$ summarizes the history before time t, $n(\cdot)$ is a feed-forward network, and $\rho_k(t)$ is the conditional probability that item k appears in the next set. When set embeddings are required as recurrent inputs, Chang et al. represent a non-empty set by the average of its item embeddings:

$$e(x) = \frac{1}{|x|} \sum_{k \in x} w_k, \qquad x \neq \emptyset,$$

and set $e(\emptyset) = 0$. This construction allows us to model simultaneous order-book events without expanding the mark space into 2^K independent labels.

We now calibrate the ground intensity $\lambda^*(t)$, denoted below by $\Lambda(t)$, using a linear Hawkes process with an exponential-mixture kernel:

$$\Lambda(t) = \mu_0 + \int_0^t \varphi(t-u)\, dN(u), \qquad \varphi(\tau) = \sum_{m=1}^{M} a_m e^{-\delta_m \tau}.$$

Here, $N(t)$ counts set-event arrivals, i.e., $dN(t)$ increments once when a subset x_i occurs, regardless of its cardinality. Equivalently, defining the decayed trace

$$s_m(t) = \sum_{t_i < t} e^{-\delta_m (t - t_i)},$$

the ground intensity can be written as $\Lambda(t) = \mu_0 + \sum_{m=1}^{M} a_m s_m(t)$.

Let $\bar{\Lambda} = \mathbb{E}[\Lambda(t)]$ denote the stationary mean ground rate. Under stationarity, $\mathbb{E}[dN(u)] = \bar{\Lambda}\, du$. Taking expectations in the Hawkes intensity equation gives the fixed-point relation

$$\bar{\Lambda} = \mu_0 + \int_0^\infty \varphi(\tau)\, \bar{\Lambda}\, d\tau = \mu_0 + \bar{\Lambda}\, n,$$

where

$$n := \int_0^\infty \varphi(\tau)\, d\tau = \sum_{m=1}^{M} a_m \int_0^\infty e^{-\delta_m \tau}\, d\tau = \sum_{m=1}^{M} \frac{a_m}{\delta_m}$$

is the branching ratio. Solving the fixed-point equation yields

$$\bar{\Lambda} = \frac{\mu_0}{1 - n}, \qquad n < 1.$$

Therefore, if we set $\mu_0 = R_{\text{target}}(1 - n)$, then

$$\bar{\Lambda} = \frac{R_{\text{target}}(1 - n)}{1 - n} = R_{\text{target}}.$$

The stationary mean ground rate is therefore pinned exactly to the target rate R_{target}, provided the process remains subcritical.

In implementation, the excitation parameters are constrained to be non-negative, e.g., $a_m = \text{softplus}(\tilde{a}_m)$, and the branching ratio is computed on each forward pass as

$$n = \sum_{m=1}^{M} \frac{\text{softplus}(\tilde{a}_m)}{\delta_m}.$$

We then set $\mu_0 = R_{\text{target}}(1 - n)$ after projecting or clipping n below one. This makes the base rate a deterministic function of the excitation mass rather than an independently learned parameter. Consequently, the optimizer can learn the temporal clustering shape through a_m/δ_m, but it cannot inflate the long-run mean event rate. Since the Chang-style conditional set distribution is normalized over subsets, the set head changes only the composition of simultaneous events and leaves the scalar Hawkes mean calibration unchanged.

Set-Valued and Single-Item Output as a Mark-Head Choice

A useful consequence of the factorization $\lambda_x^*(t) = \Lambda(t)\, p^*(x \mid t)$ is that the timing model and the calibration are *agnostic* to the support of the mark distribution $p^*(\cdot \mid t)$. The ground rate $\Lambda(t)$ counts set-event arrivals (one increment per timestamp, irrespective of cardinality), so the branching ratio n and the rate pin $\mu_0 = R_{\text{target}}(1 - n)$ are defined entirely on the arrival clock and are untouched by how the mark mass is distributed. The single-item versus set-valued distinction is therefore a localized *mark-head* choice over the support of p^*.

Write $\mathcal{X} = \mathcal{P}(\{1, \ldots, K\})$. Both heads consume the same per-type logits $z_k = z_k(h_\theta(t))$ and differ only in the final normalization and likelihood:

Single-item (categorical) head. The support is restricted to the singletons $\{k\}$, and p^* is a soft-max over types,

$$p^*(\{k\} \mid t) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad \sum_{k=1}^{K} p^*(\{k\} \mid t) = 1,$$

trained by categorical cross-entropy. Each arrival is one event type.

Set-valued (Dynamic Bernoulli) head. The support is the full power set, and p^* factorizes into independent per-type Bernoullis with $\rho_k = \sigma(z_k)$,

$$p^*(x \mid t) = \prod_{k \in x} \rho_k \prod_{k \notin x} (1 - \rho_k),$$

trained by the summed per-type Bernoulli log-likelihood. Each arrival is a subset of event types, modelling genuine simultaneity.

Both heads are rate-neutral, so calibration is preserved either way. Rate-neutrality is the statement $\sum_x p^*(x \mid t) = 1$ over the chosen support, which holds in both cases: the soft-max sums to one over the K singletons, while the product-of-Bernoullis sums to one over the entire power set,

$$\sum_{x \subseteq \{1,\ldots,K\}} \prod_{k \in x} \rho_k \prod_{k \notin x} (1 - \rho_k) = \prod_{k=1}^{K} \big(\rho_k + (1 - \rho_k)\big) = 1.$$

In either case $\sum_x \lambda_x^*(t) = \Lambda(t)$, so the stationary-mean pin $\bar{\Lambda} = R_{\text{target}}$ and the branching certificate are independent of the mark-head choice. A single event is simply the singleton $x = \{k\}$ ($|x| = 1$), so the set-valued formulation subsumes single-item output as a special case, and the multi-hot event encoding represents it natively.

The practical guidance follows directly: pick the head to match the data, not the timing model. When arrivals are effectively atomic, the categorical head is exact and economical; when simultaneity ($|x| > 1$) is material, the Dynamic Bernoulli head represents it faithfully, while the categorical head must collapse each set to a single representative type and is then a lossy approximation. Crucially, switching between the two changes only the mark head's activation and likelihood—never the ground rate, its calibration, or the set-valued input encoding.

Empirical - Into system design

Data Selection

1. heatmap dynamic pump vs non pump

We process the full orderbook data events but keep only the top 10 for orderbook modeling, which is the practice used in the industry (Shi and Cartlidge 2022).

Memory issue when processing the orderbook construction

Since USDT pairs are not available across all exchanges, when there is not USDT as the base commodity we choose the USD as the base, since the USDT is pegged to USD and the exchange rate stay 1 to 1 ratio normally.

The order book data from Kaiko are records of updates or snapshots per timestamps, but there are rare errors where duplicate timestamps exist–less than 0.1% of the overall orderbook snapshots. In this case, we save the snapshots of both duplicate timestamps but skip the event generations

According to (Cong et al. 2023) Coinbase, Gemini for regulated, normal Binance, Huobi for unregulated, normal Bitfinex, Kucoin for unregulated, abnormal

BTC highest rejection XRP second highest rejection ETH lowest reject

Most of the exchanges have multiple update per exchange, although Gemini has only one update per timestamp.

The Kaiko steam has "u" and "s" in the order book data steam, which stand for updates and snapshots. Most of the exchanges has dominate updates and a few snapshots. However, for Kucoin, all of its steams are snapshots.

- TOP MARKET CAP (6): BTC, ETH, SOL, XRP, TON, AVAX
- MAJOR L1/L2 BLOCKCHAINS (5): DOT, POL, INJ, XTZ, FET
- DEFI PROTOCOLS (8): AAVE, UNI, COMP, CRV, LDO, SNX, SUSHI, YFI
- MEME COINS (6): DOGE, SHIB, PEPE, FLOKI, BONK, WIF
- GAMING/METAVERSE (3): SAND, AXS, APE
- ORACLE/DATA INFRASTRUCTURE (4): LINK, GRT, FIL, AR
- PRIVACY/PAYMENTS (2): ZEC, LTC
- DEX/TRADING INFRASTRUCTURE (3): LRC, ZRX, SKY
- OTHER/UTILITY (2): AMP, CHZ

1. how the orderbook looks like in statistics and whats the abnormal
2. pump-and-dump period order frequency
3. wash trading exchange level order amount
4. spread big or small

System Design

!!The state transition matrix phi[K,W,W] is trained offline via simple MLE (counting transitions), completely separate from the neural network.

We illustrate the pipeline of TFOW in 1, which is composed of three key modules: (i) data collector; (ii) event constructor; (iii) anomaly detector. The data collector gathers trades and limit order book updates, which are then transformed by the event constructor into a sequence of order events. These sequences serve as inputs to the anomaly detector, which first identifies abnormal sequences using the sum-of-squared-spacings statistic, and subsequently localizes the anomalies via hitting-time and A-before-B query queries. Note, the anomaly detector starts by raising deviation alerts at the sequence level, and then pinpoints anomalies at timestamps and uncovers the underlying temporal motifs.

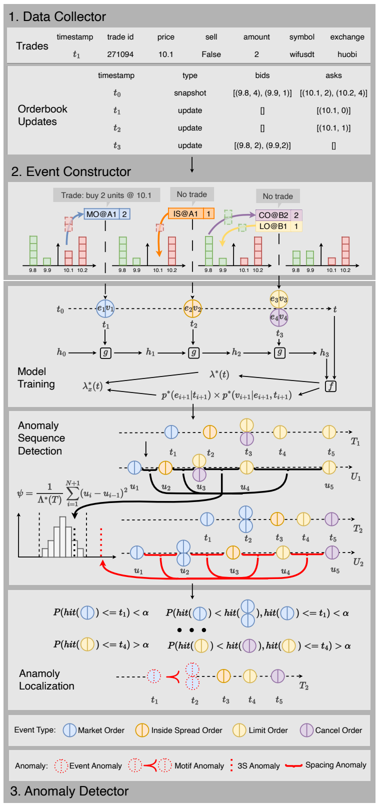

Figure 1: TFOW consists of three modules: (i) data collector; (ii) Event constructor; and (iii) anomaly detector. Note, (a, b) here is not an event but the actual price and depth.

Data Collector

We obtain both trade and Level-2 limit order book data from Kaiko (Kaiko 2025), a leading institutional provider of digital asset market data. In partnership with Cloudburst Technologies (Cloudburst 2025), a leading firm specializing in detecting and mitigating market manipulation, we establish the data collector. Level-2 limit order book updates are continuously streamed to a Google Cloud Storage bucket, providing full-depth market coverage. Each update record contains precise timestamps, update types (snapshots or liquidity updates), and the corresponding price–volume information for bid and ask levels. Complementary trade data are retrieved via Kaiko's API and temporally aligned with the order book stream, enabling synchronized analysis of trading activity and order flow dynamics.

Event constructor

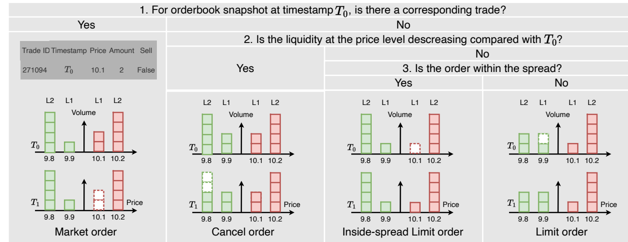

Figure 2: Event construction for 4 order types through 3 questions. Each question is a decision tree for classification. We start with question 1 to question 3. Note, market order is a case of liquidity decreasing.

To apply point process modeling to high-frequency limit order book (LOB) data, we represent market dynamics as a sequence of **event sets**. We consider the top $K = 10$ levels on each side (bid and ask).

Atomic Events. We define a finite set E of *atomic event types*, each denoted $e = (c, s, \delta)$, where:

- $c \in \{\mathrm{LO}, \mathrm{MO}, \mathrm{CO}, \mathrm{IS}\}$ is the *event class*:
- **Limit Order (LO)**: liquidity provision at one of the top-K levels.
- **Cancel Order (CO)**: liquidity withdrawal at one of the top-K levels.
- **Market Order (MO)**: liquidity consumption at the best quote.
- **Inside-Spread Order (IS)**: a new order placed inside the spread, improving the prevailing best quote.
- $s \in \{b, a\}$ is the *side* (bid or ask).
- δ is the *level index*:
- For LO/CO, $\delta \in \{1, \ldots, K\}$ indicates distance from the best quote ($\delta = 1$ = best bid/ask, $\delta = 10$ = tenth level).
- For MO, $\delta = 1$ always, since market orders execute against the current best quote.
- For IS, $\delta \in \{1, \ldots, K\}$ denotes the number of ticks by which the new quote improves the spread (e.g., $\mathrm{IS}_{b,2}$ = new bid order two ticks inside the spread).

Event Construction from LOB and Trade Data. As illustrated in 2, each atomic order event is inferred by differencing consecutive Level-2 order book snapshots and aligning them with trade records at identical timestamps. Because trades and book updates are perfectly synchronized in both time and volume, every observed change can be classified without ambiguity through a three-step decision tree. First, if the best-quote liquidity decreases and a trade is recorded at the same price, the change corresponds to a **Market Order**—a direct execution against standing quotes. Second, if liquidity decreases at any level without a matching trade, the reduction represents a **Cancel Order**, reflecting liquidity withdrawal. Finally, when liquidity increases, the classification depends on the price position: if the new order appears inside the spread, it forms an **Inside-Spread Order**; otherwise, it is a **Limit Order** placed within the existing book depth. This rule set ensures exhaustive mapping from raw order book and trade data into atomic order events, enabling a consistent foundation for downstream temporal modeling and anomaly detection.

Event Sets. In crypto limit order books, each event is associated with a set of items. In other words, multiple atomic events may occur simultaneously at the same timestamp. We therefore allow *event sets* at each time t_i:

$$X_{t_i} = \{(e_{i_1}, v(e_{i_1})), \dots, (e_{i_n}, v(e_{i_n}))\},$$

where $e_{i_n} \in \mathcal{E}$ are atomic events and $v(e_{i_n})$ their volume marks.

State-Dependent Set-MTPP Backbone

Intuition. The event flow of a limit order book is governed not only by past order-flow history, but also by the contemporaneous market condition. In particular, the relative dominance of bid-side and ask-side queues affects the likelihood of future submissions, cancellations, executions, and inside-spread improvements. Following the state-dependent neural Hawkes formulation of Shi and Cartlidge, we incorporate a discrete market state into the intensity model so that TFOW captures both temporal dependence and the current order-book regime. Unlike Shi and Cartlidge, however, we do *not* adopt the parallel CT-LSTM architecture. We only use their state mechanism and retain a single continuous-time neural backbone, which is more suitable for our set-valued marked event representation and keeps the model compact.

Market state. Let $v_b^{(1)}(t)$ and $v_a^{(1)}(t)$ denote the top-of-book bid and ask queue volumes at time t. We define the queue-imbalance indicator

$$I(t) = \frac{v_b^{(1)}(t) - v_a^{(1)}(t)}{v_b^{(1)}(t) + v_a^{(1)}(t)}.$$

Given a threshold $\theta \in (0, 1)$, we discretize the market state as

$$s_t = \begin{cases} 0, & I(t) \in [-1, -\theta), \\ 1, & I(t) \in [-\theta, \theta], \\ 2, & I(t) \in (\theta, 1]. \end{cases}$$

These three regimes correspond to ask-dominant, balanced, and bid-dominant markets, respectively.

State-dependent hidden dynamics. Let $\mathcal{H}_t$ denote the history of all event sets and volume marks up to time t. As defined in Section 4.2, each timestamp t_i is associated with an event set

$$X_{t_i} = \{(e_{i1}, v(e_{i1})), \dots, (e_{in_i}, v(e_{in_i}))\},$$

where $e_{im} \in \mathcal{E}$ is an atomic event and $v(e_{im})$ is its volume mark. TFOW encodes the sequence of past event sets together with the current market state into a continuous-time hidden representation h_t. Between event times, the hidden state decays continuously; at each event time, it is updated using the newly observed event set and the contemporaneous state. This follows the state-dependent neural point process principle that the intensity is decoded from a latent state rather than specified by an additive Hawkes kernel.

We define the baseline intensity by

$$\lambda_\theta^{\mathrm{base}}(t \mid \mathcal{H}_t, s_t) = f_\theta(h_t, s_t),$$

where $f_\theta(\cdot)$ is a positive decoder. In practice, f_θ may be implemented using a softplus output layer to ensure non-negativity of the intensity.

State transition. The market state is allowed to evolve jointly with the event process. Following Shi and Cartlidge, after the event set at time t_i is observed, the next state is governed by an event-conditioned transition distribution

$$P(s_{t_i} \mid \mathcal{H}_{t_i^-}, y_i, s_{t_i^-}) \sim \phi_{y_i, s_{t_i^-}},$$

where y_i is the event-type summary associated with timestamp t_i, and $\phi \in \mathbb{R}^{K \times W \times W}$ is a discrete transition tensor. In our implementation, ϕ is estimated offline via empirical transition counting, separately from the neural network parameters.

Compensator. The compensator associated with the baseline state-dependent intensity is

$$\Lambda_\theta^{\mathrm{base}}(t) = \int_0^t \lambda_\theta^{\mathrm{base}}(u \mid \mathcal{H}_u, s_u)\, du.$$

This baseline Set-MTPP models the structural dynamics of normal order flow. The next subsection introduces a second-order diagnostic derived from compensator increments and uses it to adapt the baseline intensity online.

Differential 3S as a Second-Order Diagnostic

Intuition. A correctly specified temporal point process becomes a unit-rate Poisson process after the random time change induced by its compensator. Therefore, on the transformed clock, the inter-event gaps should behave like i.i.d. $\mathrm{Exp}(1)$ random variables. This implies not only that the model gets the expected event count right, but also that it gets the geometry of event spacing right. In particular, if $\Delta u_i \sim \mathrm{Exp}(1)$, then

$$\mathbb{E}[\Delta u_i] = 1, \qquad \mathrm{Var}(\Delta u_i) = 1, \qquad \mathbb{E}[(\Delta u_i)^2] = 2.$$

The original 3S statistic is built from squared transformed gaps and therefore probes this second-moment structure. Intuitively, first-order residuals ask whether the model gets the *number of arrivals* right, while squared compensator increments ask whether it gets the *irregularity of event timing* right. This motivates a sequential second-order diagnostic for TFOW.

Compensator increments. Let

$$u_i = \Lambda_\theta^{\mathrm{base}}(t_i), \qquad \Delta u_i = u_i - u_{i-1} = \int_{t_{i-1}}^{t_i} \lambda_\theta^{\mathrm{base}}(u \mid \mathcal{H}_u, s_u)\, du.$$

If the baseline state-dependent model is correctly specified, then by the random time-change theorem,

$$\Delta u_i \overset{i.i.d.}{\sim} \mathrm{Exp}(1).$$

Hence,

$$\mathbb{E}[(\Delta u_i)^2] = 2.$$

This second moment is the key quantity behind 3S. Rather than computing a global goodness-of-fit statistic over an entire transformed sequence, we extract an event-level residual from the same quadratic intuition.

Differential 3S residual. We define the differential 3S residual by

$$r_i^{(2)} = (\Delta u_i)^2 - 2.$$

Under the correctly specified model,

$$\mathbb{E}[r_i^{(2)}] = 0.$$

Thus, $r_i^{(2)}$ measures the local second-order deviation of transformed spacing from the Poisson benchmark. Positive values indicate abnormally irregular, bursty, or dispersed spacing on the compensator clock, whereas negative values indicate overly regular spacing. In this sense, $r_i^{(2)}$ is a quadratic second-order diagnostic on compensator increments.

Smoothed deviation state. To obtain a stable sequential signal, we smooth the event-level residual recursively:

$$d_i^{(2)} = \rho d_{i-1}^{(2)} + (1-\rho) r_i^{(2)}, \qquad \rho \in [0,1).$$

The process $d_i^{(2)}$ summarizes recent second-order deviations in transformed event spacing. Unlike a rolling-window goodness-of-fit test, this construction is incremental and naturally compatible with sequential modeling.

Deviation-controlled intensity. We incorporate the second-order deviation signal into TFOW through a multiplicative correction of the baseline state-dependent intensity:

$$\lambda_\theta(t \mid \mathcal{H}_t, s_t) = \lambda_\theta^{\text{base}}(t \mid \mathcal{H}_t, s_t) \exp\big(\beta d_i^{(2)}\big), \qquad t \in (t_i, t_{i+1}],$$

where $\beta \in \mathbb{R}$ is learnable. Equivalently,

$$\log \lambda_\theta(t \mid \mathcal{H}_t, s_t) = \log \lambda_\theta^{\text{base}}(t \mid \mathcal{H}_t, s_t) + \beta d_i^{(2)}.$$

This decomposition separates structural order-flow dynamics from second-order temporal irregularity: the baseline term captures the dynamics explained by event history and imbalance state, while the differential 3S term captures residual spacing abnormalities not explained by the structural model.

Discussion. The proposed diagnostic is complementary to first-order residual monitoring. Standard martingale residuals compare realized and expected counts and are therefore sensitive to count drift. In contrast, differential 3S probes the second moment of transformed waiting times and is sensitive to clustering, dispersion, and abnormal spacing geometry. In crypto limit order books, this allows TFOW to respond not only to unusual event volume, but also to strategically timed bursts, pauses, and irregular cancellation or inside-spread patterns that are characteristic of manipulative flow.

Differential 3S as a Second-Order Diagnostic on Compensator Increments

State-Dependent Intensity Modeling In limit order book markets, the arrival of events is influenced not only by past order flow but also by the contemporaneous market condition. Recent work therefore extends neural Hawkes processes by conditioning the event intensity on a discrete market state that summarizes the current order book configuration (Shi and Cartlidge 2022).

Formally, let H_t denote the event history and s_t a state variable derived from the order book. The conditional intensity for event type k is then modeled as

$$\lambda_k(t \mid H_t, s_t) = f_k(h_t, s_t), \tag{1}$$

where h_t is a latent representation of past events produced by a continuous-time recurrent mechanism and $f_k(\cdot)$ decodes the instantaneous arrival rate. The state variable typically partitions the market into regimes such as bid-dominant, balanced, or ask-dominant based on queue imbalance.

We extend the intensity formulation by introducing a deviation-controlled feedback mechanism that enables explicit separation between structural dynamics and market-driven temporal irregularity.

Baseline TFOW intensity. Model's conditional intensity is

$$\lambda_0(t) = \lambda(h_t, x_t), \tag{2}$$

where h_t encodes historical order flow and x_t denotes the imbalance regime state. The associated compensator is

$$\Lambda_0(t) = \int_0^t \lambda_0(\tau)\, d\tau. \tag{3}$$

The derivation of $\lambda_0(t)$ and $\Lambda_0(t)$ follows the temporal point process.

$$\Delta u_i = \Lambda(t_i) - \Lambda(t_{i-1}) = \int_{t_{i-1}}^{t_i} \lambda(\tau)\, d\tau. \tag{4}$$

Under correct model specification, the time-rescaling theorem implies

$$\Delta u_i \sim \text{Exp}(1),$$

i.e., rescaled inter-arrival times are i.i.d. exponential with unit mean.

Intuition. A correctly specified temporal point process should become a unit-rate Poisson process after the random time change induced by its compensator. On this transformed clock, the inter-event gaps should behave like i.i.d. $\text{Exp}(1)$ variables. This implies not only a correct *first-order* behavior, i.e., the expected number of arrivals matches the compensator, but also a correct *second-order* behavior, i.e., the transformed waiting times have the right dispersion and spacing geometry. In particular, if $\Delta u_i \sim \text{Exp}(1)$, then

$$\mathbb{E}[\Delta u_i] = 1, \qquad \text{Var}(\Delta u_i) = 1, \qquad \mathbb{E}[(\Delta u_i)^2] = 2.$$

The classical 3S statistic is built from squared transformed gaps and therefore probes this second-moment structure.

Intuitively, while first-order residuals measure whether the model gets the *event count* right, squared compensator increments measure whether the model gets the *timing irregularity* right. Large values indicate abnormally bursty or dispersed spacing on the transformed clock, while small values indicate overly regular spacing. This makes the second moment of compensator increments a natural second-order diagnostic for local model misspecification.

Formally, let $\lambda_\theta(t \mid \mathcal{H}_t)$ denote the conditional intensity produced by TFOW, and define its compensator

$$\Lambda_\theta(t) = \int_0^t \lambda_\theta(u \mid \mathcal{H}_u)\, du.$$

For the observed event times $\{t_i\}_{i=1}^N$, define the transformed times

$$u_i = \Lambda_\theta(t_i), \qquad \Delta u_i = u_i - u_{i-1}.$$

By the random time-change theorem, if the model is correctly specified, then the transformed process is a standard Poisson process and the increments satisfy

$$\Delta u_i \overset{i.i.d.}{\sim} \mathrm{Exp}(1).$$

Motivated by the quadratic structure of 3S, we define the event-level *differential 3S residual*

$$r_i^{(2)} = (\Delta u_i)^2 - 2.$$

Since $\mathbb{E}[(\Delta u_i)^2] = 2$ under the Poisson null, $r_i^{(2)}$ is centered:

$$\mathbb{E}[r_i^{(2)}] = 0.$$

Thus, $r_i^{(2)}$ quantifies the local second-order deviation of the transformed spacing from the model-implied $\mathrm{Exp}(1)$ benchmark. Specifically, $r_i^{(2)} > 0$ indicates abnormally irregular transformed spacing, whereas $r_i^{(2)} < 0$ indicates overly regular spacing.

To obtain a stable sequential signal, we smooth the event-level residual recursively:

$$d_i^{(2)} = \rho d_{i-1}^{(2)} + (1-\rho) r_i^{(2)}, \qquad \rho \in [0, 1).$$

The state $d_i^{(2)}$ summarizes recent second-order deviations on the compensator clock. We then incorporate it into the conditional intensity as a multiplicative correction:

$$\lambda_\theta(t \mid \mathcal{H}_t) = \lambda_\theta^{\mathrm{base}}(t \mid \mathcal{H}_t) \exp\bigl(\beta d_i^{(2)}\bigr), \qquad t \in (t_i, t_{i+1}],$$

where $\lambda_\theta^{\mathrm{base}}$ is the base TFOW intensity and β controls the sensitivity to second-order deviation.

This construction is closely related in spirit to 3S, since both are driven by squared transformed inter-event gaps. The distinction is that the original 3S statistic is a *global* goodness-of-fit functional, whereas our formulation extracts an *incremental* second-order residual process suitable for sequential modeling. Consequently, TFOW is able to respond online not only to the structural order-flow state, but also to recent deviations in the timing geometry of events relative to the compensator-implied Poisson benchmark.

Discussion. The proposed diagnostic is complementary to first-order residual monitoring. Standard martingale residuals compare observed and expected counts through $N(t) - \Lambda_\theta(t)$, and are therefore sensitive to count drift. In contrast, $r_i^{(2)}$ probes the second moment of transformed waiting times and is sensitive to local irregularity in event spacing. In microstructure settings, this allows the model to detect regime shifts that manifest not only through excess event volume, but also through abnormal clustering, dispersion, or regularization of order-flow timing.

State-Dependent Intensity with Deviation Feedback

Deviation-controlled intensity. We define the modified intensity:

$$\boxed{\lambda(t) = \lambda_0(t) \cdot \exp(\beta s_t),} \tag{5}$$

with $\beta \in \mathbb{R}$ learnable.

Taking logarithms,

$$\log \lambda(t) = \log \lambda_0(t) + \beta s_t. \tag{6}$$

The baseline term $\lambda_0(t)$ captures structural dynamics encoded in (h_t, x_t), which summarize event history and regime information. Any deviation in arrival behavior not explained by these structural components must manifest through systematic distortions of rescaled-time spacings.

Since s_t is computed solely from compensator increments $\{\Delta u_i\}$, it measures inconsistency between predicted and realized temporal spacing. Importantly, s_t depends only on the temporal arrangement of events relative to the model's own predicted intensity and under correct structural modeling, fluctuations in s_t arise from genuine clustering or dispersion in the market. Persistent elevation of s_t corresponds to bursty arrivals (volatility shocks, liquidation cascades), and suppressed s_t corresponds to under-dispersed flow (liquidity drought, inactivity).

Therefore, βs_t quantifies multiplicative amplification (or attenuation) of structural intensity required to reconcile predicted flow with observed temporal irregularity.

In other words,

$$\beta s_t = \log \frac{\lambda(t)}{\lambda_0(t)}, \tag{7}$$

which measures the deviation of realized market activity from structurally expected activity.

Let $\hat{\lambda}_{\mathrm{obs}}(t)$ denote empirical intensity estimated over a short window. We define

$$D_{\mathrm{market}}(t) = \beta s_t, \tag{8}$$

$$D_{\mathrm{model}}(t) = \left| \log \hat{\lambda}_{\mathrm{obs}}(t) - \log \lambda(t) \right|. \tag{9}$$

Then

$$\log \hat{\lambda}_{\mathrm{obs}}(t) = \log \lambda_0(t) + D_{\mathrm{market}}(t) + \epsilon_t, \tag{10}$$

where ϵ_t corresponds to residual model misspecification.

Interpretational summary.

- $\lambda_0(t)$: structural prediction from TFOW history and regime.
- βs_t: systematic temporal irregularity induced by market dynamics.
- ϵ_t: unexplained residual error.

Thus, βs_t represents market-driven deviation because it quantifies the multiplicative adjustment necessary to align structural intensity with empirically observed clustering, independently of latent state encoding.

This decomposition renders temporal abnormality identifiable and enables real-time attribution of intensity variation to market shocks rather than model error.

Model Training !!!* train baseline intensity first * compute differential 3S from detached compensator increments * feed it back as a correction feature

Modeling Event Type and Volume in Order book Data The event constructor outputs a sequence of set of events. Formally, let each event at time t_i be represented as a tuple

$$x_i = (e_i, v_i),$$

where e is the event type and $v_i \in \mathbb{R}^+$ is the corresponding volume. We factorize the conditional mark distribution as

$$p(x_i \mid t_i, H(t_i)) = p(e_i \mid t_i, H(t_i)) \cdot p(v_i \mid e_i, t_i, H(t_i)), \tag{11}$$

where $H(t_i)$ denotes the event history up to time t_i.

Set-Valued Marked Temporal Point Processes (Set-MTPP) Classical marked temporal point processes (MTPPs) assume that each event is associated with a single categorical mark. This simplification breaks down in crypto limit order book level-2 data nature, where multiple atomic events—such as simultaneous limit placements, cancellations, and executions—often occur at exactly the same timestamp. To properly capture these co-occurrences, we adopt the *set-valued MTPP* framework (Chang, Boyd, and Smyth 2024b), which generalizes neural MTPPs from singleton events to event *sets*.

Formally, let the event history be

$$H(T) = \{(\tau_i, X_i)\}_{i=1}^{N},$$

where $\tau_i \in \mathbb{R}^+$ is the i-th event time and $X_i \subseteq \{1, \ldots, M\}$ is a finite subset of atomic event type e . The history up to time t is $H(t) = \{(\tau_j, X_j) : \tau_j < t\}$.

The process is parameterized by a ground intensity $\lambda^*(t)$ and a conditional distribution over sets $p^*(X \mid t, H(t))$. The marked intensity factorizes as

$$\lambda^*_X(t) = \lambda^*(t)\, p^*(X \mid t, H(t)).$$

Accordingly, the joint log-likelihood decomposes into temporal and set contributions:

$$\mathcal{L}(H(T)) = \underbrace{\sum_{i=1}^{N} \log \lambda^*(\tau_i) - \int_0^T \lambda^*(s)\, ds}_{\mathcal{L}_{\text{time}}} + \underbrace{\sum_{i=1}^{N} \log p^*(X_i \mid \tau_i, H(\tau_i))}_{\mathcal{L}_{\text{set}}}$$

Set-MTPPs therefore provide a principled generative framework for level-2 LOB modeling, separating the dynamics of *when* events occur from *which sets* of events occur. This enables likelihood-based inference over simultaneous market microstructural patterns.

Volume Set Multivariate Temporal Point Processes While set-MTPPs capture simultaneous event types, each LOB event also carries a continuous *volume mark* that reflects liquidity provision or consumption. Volumes are heavy-tailed and constrained by available depth—features that categorical marks alone cannot capture. We therefore extend set-MTPPs to a *volume-set-MTPP*, which models both the composition of event sets and their associated traded volumes under realistic market constraints.

At each event time t_i, we observe:

$$X_i \subseteq \{1, \ldots, M\}, \quad \mathbf{v}_{t_i} = \{v(e) \in \mathbb{R}_{>0} : e \in X_{t_i}\},$$

where X_{t_i} is the set of event types and $\mathbf{v}_{t_i}$ their associated volumes. The modeling task is to learn

$$p(X_t, \mathbf{v}_t \mid H_t),$$

where H_t is the market history up to t.

Set–Volume Factorization. We decouple set composition and volume magnitudes:

$$p(X_t, \mathbf{v}_t \mid H_t) = p(X_t \mid H_t) \prod_{e \in X_t} p(v(e) \mid e, H_t, \text{cap}_t(e)).$$

This allows the model to (i) capture co-occurrence structure in $p(X_t \mid H_t)$, (ii) model event-specific volume distributions, and (iii) respect market microstructure constraints via $\text{cap}_t(e)$.

Dynamic Set Model. Event sets are modeled via conditionally independent Bernoulli variables:

$$p(X_t = x \mid H_t) = \prod_{k=1}^{K} \rho_k(h_t)^{x_k} (1 - \rho_k(h_t))^{1 - x_k},$$

with $\rho_k(h_t) = \sigma(w_k^\top n(h_t) + b_k)$ parameterized by the hidden state h_t.

Neural Architecture and Training. We employ a Neural Hawkes (Mei and Eisner 2017) to produce hidden states $h_{d,t}$, with three prediction heads: (i) an intensity head for timing $\lambda(t)$, (ii) a set head for $\{\rho_k(h_{d,t})\}$, and (iii) a volume head for $\{(\mu_k(h_{d,t}), \sigma_k(h_{d,t}))\}$. The loss decomposes as

$$\mathcal{L} = \mathcal{L}_{\text{timing}} + \mathcal{L}_{\text{sets}} + \mathcal{L}_{\text{volumes}},$$

balancing temporal, set, and volume likelihood terms.

This formulation yields a principled and high-fidelity generative model of LOB microstructure—enabling both simulation and anomaly detection in high-frequency trading.

Anomaly Detection

Market manipulators deliberately inject crafted orders into otherwise normal order flows, perturbing the natural distribution of event timings, traded volumes, and directional dependencies between orders. Such distortions are inherently statistical and can therefore be detected through distributional tests on the reconstructed event history.

Formally, let the full observed history up to horizon T be

$$\mathcal{H}(T) = \{(t_i, X_{t_i})\}_{i=1}^{N},$$

where $t_1 < t_2 < \cdots < t_N \leq T$ are event times and X_{t_i} denotes the event set observed at time t_i.

Stage 1: Sequence-Level Detection. We first segment the full history into hourly subsequences and test each for deviations from the learned market baseline. Specifically, we define

$$\mathcal{H}_i = \mathcal{H}[t_a, t_b] = \{(t_j, X_{t_j}) : t_a < t_j \leq t_b\}, \qquad D_{\text{train}} = \{\mathcal{H}_1, \ldots, \mathcal{H}_M\},$$

where the subsequences in D_{train} are assumed to follow the empirical distribution P_{data}. For every $\mathcal{H}_i$, we conduct the hypothesis test

$$H_0 : \mathcal{H}_i \sim P_{\text{data}} \quad \text{vs.} \quad H_1 : \mathcal{H}_i \sim Q,\ Q \neq P_{\text{data}}.$$

To quantify deviations, we apply the *sum-of-squared-spacings (3S)* statistic:

$$\psi(Z) = \tfrac{1}{V} \sum_{i=1}^{N+1} (v_i - v_{i-1})^2, \qquad Z = (\Lambda^*(t_1), \ldots, \Lambda^*(t_N)),$$

where $\Lambda^*(t)$ is the fitted compensator of the model intensity and $Z = \Lambda^*(T)$ the total rescaled horizon. The 3S statistic simultaneously captures irregularities in event spacing and cumulative mass flow. Subsequences with low p-values under P_{data} are flagged as anomalous trading hours.

Stage 2: Event-Level Localization. For each detected anomalous window $\mathcal{H}[a_t, b_t]$, TFOW proceeds to *localize* the responsible atomic events through probabilistic queries derived from the fitted process.

(a) Timestamp-level test. We first evaluate *hitting-time anomalies*—rare early arrivals of specific event types—by computing

$$\mathbb{P}\big(\text{hit}(A) < t \,\big|\, \mathcal{F}_{t_a}\big), \qquad \mathcal{F}_{t_a} = \mathcal{H}[0, t_a) = \{(t_j, X_{t_j}) : t_j < t_a\}.$$

Significantly smaller values than expected under P_{data} indicate abnormally early occurrences of events in subset A, pointing to potential manipulative bursts.

(b) Motif-level test. When such deviations are present, we escalate to *motif-level localization*, focusing on the minimal temporal motif between two consecutive timestamps $(t_{i-1}, t_i]$. For disjoint event subsets A, B, we compute the directional probability

$$q(A \prec B) = \Pr(t_A < t_B \mid P_{\text{data}}),$$

which measures how often A precedes B under the learned dynamics. Anomalously small $q(A \prec B)$ values expose rare directional orderings—interpretable microstructural signatures of manipulative flow.

This two-stage design ensures a consistent "small-is-anomalous" interpretation: the 3S test surfaces statistically irregular trading intervals; hitting-time queries highlight unexpectedly early local arrivals; and motif queries explain directional distortions between consecutive event sets. Together they yield both *statistical validity* and *microstructural interpretability*, enabling TFOW to pinpoint manipulative behaviors at order-level granularity.

Anomaly Sequence Detection We extend the probabilistic testing paradigm of Shchur et al. (Shchur et al. 2021a) to the limit order book setting with set-valued events and continuous volumes. The central idea is to map observed event sequences into the standard Poisson process (SPP) via appropriate compensators, and then apply *sum-of-squared-spacings* (3S) tests. This unifies the detection of anomalies in timing, spacing, and traded volumes.

Sum-of-Squared-Spacings. We adopt the sum-of-squared-spacings (3S) statistic for order book anomaly detection. Given fitted total intensity

$$\lambda^*(t) = \sum_{x \in \mathcal{X}} \lambda_x^*(t)$$

, the compensator

$$\Lambda^*(t) = \sum_{x \in \mathcal{X}} \Lambda_x^*(t) = \sum_{x \in \mathcal{X}} \int_0^t \lambda_x^*(u)\, du$$

maps arrivals t_i to rescaled times $u_i = \Lambda^*(t_i)$. The statistic

$$\psi = \frac{1}{\Lambda^*(T)} \sum_{i=1}^{N+1} (u_i - u_{i-1})^2$$

quantifies spacing irregularities and count deviations simultaneously. Unlike KS-based tests, 3S captures bursts, lulls, and atypical event counts; when extended with volume compensators, it further detects abnormal liquidity flows, making it well suited for identifying manipulative order flow.

Mass-Flow Deviations. Manipulation often manifests not in counts but in *abnormal volume flow*. We define the *mass intensity*

$$\rho^*(t) = \sum_{k=1}^{K} \lambda_k^*(t)\, \mathbb{E}[v \mid k, t, H_t], \qquad \Xi^*(t) = \int_0^t \rho^*(u)\, du,$$

where $\mathbb{E}[v \mid k, t, H_t]$ is the truncated log-normal expectation from our volume head. Mapping arrivals to $m_i = \Xi^*(t_i)$ yields the mass-based statistic

$$\psi_{\text{mass}} = \frac{1}{\Xi^*(T)} \sum_{i=1}^{N+1} (m_i - m_{i-1})^2,$$

which flags anomalies where mass accumulates atypically.

Anomaly Localization. To achieve both statistical rigor and interpretability, we employ a two-stage anomaly localization. First, the joint 3S detector is applied to *hourly subsequences*, producing anomaly signals at the scale of trading hours. Second, within each anomalous hourly subsequence, we conduct *timestamp-level query*: for every timestamp t_i, we compute the conditional probability of its set given the immediately preceding timestamp under the fitted volume-set-MTPP. If this probability falls below a threshold, α, we escalate to the motif query over (t_{i-1}, t_i). By ranking anomalous motif $A \prec B$, we retain the top-N most anomalous consecutive sets as the *anomaly motifs*. These motifs reveal interpretable precursors of anomalies.

Hitting Time Queries. Let $A \subseteq \{1, \dots, M\}$ be a non-empty subset of atomic events. The *hitting time* $\text{hit}(A)$ is defined as the first occurrence time τ such that $X_\tau \cap A \neq \emptyset$. The probability that any event in A occurs before a horizon t can be expressed as

$$\mathbb{P}(\text{hit}(A) \le t) = 1 - \mathbb{E}_{H[0,t]\sim q}\left[\exp\Big(-\int_0^t \lambda^*(s)\, p^*(A \mid s, H(s))\, ds\Big)\right]$$

where $\lambda^*(t)$ is the ground intensity and $p^*(A \mid t, H(t))$ is the conditional probability that the sampled set at time t contains at least one item from A. For the Dynamic Bernoulli factorization, this reduces to

$$p^*(A \mid t, H(t)) = 1 - \prod_{k \in A}\big(1 - p^*(k \mid t, H(t))\big).$$

This formulation allows efficient Monte Carlo estimation under importance sampling and provides a natural way to quantify the likelihood of early occurrences, which we later use for identifying anomalously early event clusters.

A-before-B Queries. For two disjoint subsets $A, B \subseteq \{1, \dots, M\}$, we are interested in the probability that some event in A occurs before any event in B within horizon t, i.e.,

$$\mathbb{P}(\text{hit}(A) < \text{hit}(B),\, \text{hit}(A) \le t).$$

Using the factorization $\lambda^*_X(t) = \lambda^*(t)\, p^*(X \mid t, H(t))$ and importance sampling with proposal intensity restricted to $(A \cup B)$, the unbiased estimator is

$$\begin{aligned} &p(\text{hit}(A) < \text{hit}(B), \text{hit}(A) \le t) \\ &= \mathbb{E}^q_{H[0,t]}\left[\int_0^t \exp\left(-\int_0^s \lambda^*_{A\cup B}(s')\, ds'\right) \lambda^*_{A\cap\neg B}(s)\, ds\right], \end{aligned} \tag{12}$$

where $\lambda^*_S(t) = \lambda^*(t)\, p^*(S \mid t, H(t))$ for any subset S. The term $p^*(A \cap \neg B \mid t, H(t))$ represents the probability that the sampled set at t contains an item in A but none in B. Symmetric forms yield estimators for the other three cases—B before A, simultaneous occurrence, and neither before t. These directional queries form the basis for detecting causal or lead-lag relations between distinct event types in continuous time.

Localization Procedure Formally speaking, given a detected subsequence $[t_0, t_n]$ and history filtration $\{\mathcal{F}_t\}$, we test anomalies in two escalating stages. First, for $i = 1, \dots, n$, conditioning on $\mathcal{F}_{t_{i-1}}$, we compute the local hazard of hitting $A \subseteq \{1, \dots, K\}$ on $(t_{i-1}, t_i]$, with compensator $\Lambda^*_A(t_{i-1}, t_i) = \int_{t_{i-1}}^{t_i} \lambda^*(s) p^*(A \mid s, H(s))\, ds$. The one-sided p-value $p_i^{\text{hit}}(A) = \exp(-\Lambda^*_A(t_{i-1}, t_i))$ is small when A appears earlier than expected, flagging rare local hits. Second, if $p_i^{\text{hit}}(A) < \alpha_1$, we estimate the conditional probability that A precedes disjoint B in $(t_{i-1}, t_i]$:

$$q_i(A \prec B) = \int_{t_{i-1}}^{t_i} \exp\Big(-\int_{t_{i-1}}^{s} \lambda^*_{A\cup B}(u)\, du\Big)\, \lambda^*_{A\cap\neg B}(s)\, ds.$$

Here, small $q_i(A \prec B)$ values indicate that the motif $A \prec B$ is rare under the model, and thus anomalous. We therefore label $(t_{i-1}, t_i]$ as anomalous and record motif $(A \prec B)$ iff $p_i^{\text{hit}}(A) < \alpha_1$ and $q_i(A \prec B) < \alpha_2$. This construction ensures a consistent "small-is-anomalous" interpretation across both stages, yielding motifs that are simultaneously surprising in timing and rare in directional ordering.

This framework provides a statistically principled path from *detection* to *localization* to *explanation*. The 3S statistic unifies timing, spacing, and mass anomalies, but it cannot be reliably applied to sequences with very few timestamps, since the variance of the spacing distribution is ill-defined in such regimes. Our query-based refinement overcomes this limitation by leveraging probabilistic set-MTPP queries, ensuring robustness in sparse cases while extracting interpretable temporal motifs. Together, the framework delivers a high-precision, label-free anomaly detector tailored to the structure of high-frequency financial markets with real-world interpretability.

Evaluation

We evaluate the volume-set MTPP on three axes that mirror its intended role in TFOW: (i) one-step predictive quality of the next event set, its timing, and its type, against adapted neural and non-parametric baselines (Sec. –); (ii) closed-loop stability of the learned dynamics and generative realism of free-running simulations under the stylized facts of high-frequency order flow (Sec.); and (iii) price-level realism of simulated books against recorded ground truth (Sec.). Together these establish the model as a faithful generative model of LOB microstructure—the property the 3S detection and query-localization stages rely on.

Evaluation Setup

Data. We use Level-2 order book and trade data for BTC/USDT, ETH/USDT and SOL/USDT on Binance (Kaiko feed), reconstructed into atomic events by the event constructor of Section 3 (MO/CO/LO/IS $\times$ side $\times$ top K=10 levels; 62 fixed event types). The corpus covers seven consecutive trading days (1–7 January 2026) for each of the three pairs (21 daily files). Sliding windows of 50 events (stride 32) are split chronologically *within each asset* 70/15/15 into train/validation/test, with a one-window gap at each boundary to prevent leakage. Timestamps are exchange-stamped on a 0.1 s grid; the mean target set contains 3.94 of 62 types, confirming that simultaneous events are pervasive.

Models. We train the volume-set MTPP with three interchangeable continuous-time decoders: a Neural-Hawkes cell (Mei and Eisner 2017) (VSM-HAWKES), an RMTPP-style recurrent decoder (Du et al. 2016) (VSM-RMTPP), and a structured state-space decoder (VSM-S2P2). Baselines follow the borrowed-backbone protocol of Shi and

Cartlidge (Shi and Cartlidge 2022): an LSTM, a causal Transformer (SAHP-style (Zhang et al. 2020)), a decayed-cell recurrent network (CT-LSTM-style (Mei and Eisner 2017)), and a per-type parallel recurrent network (PCT-LSTM-style (Shi and Cartlidge 2022)), each fitted with identical set/volume/time heads, multi-hot event-set histories, and the same training budget (30 epochs, batch 512, AdamW, 10^{-3}). Our models carry an explicit per-channel volume head (log-volume Normal) whose negative log-likelihood over active channels enters the joint objective.[1] A non-parametric empirical transition baseline (EMPTRANS; next-set frequencies conditioned on the last event set) anchors the tables. Results are from a single seed; a three-seed extension is in progress.

Protocol. All headline metrics are *target-free*: no model receives the ground-truth next-event time. For the volume-set MTPP, the predictive distribution over the next arrival is obtained from the learned ground intensity by trapezoid quadrature of the compensator $\Lambda(t) = \int_0^t \lambda(s)\,ds$ on a log-spaced grid; point predictions read out the decision-theoretically matched statistic (predictive median for MAE, predictive mean for RMSE), and event-set probabilities are evaluated at the model-predicted time. Set decoding uses either a global threshold calibrated for validation F_1 (shared with all baselines) or a per-sample expected-F_1 rule, $k^* = \arg\max_k 2\sum_{i\le k} p_{(i)}/(k+\sum_j p_j)$ (Ye et al. 2012). As a control we also report the time-conditioned ("known-time") variant; the model's advantage does not derive from timing information.

Event-Set Prediction

Table 1 compares set-valued next-event prediction across all models under both decodes. The state-space variant (VSM-S2P2 with expected-F_1 decoding) leads every model on both per-sample Jaccard (0.202, an 8% relative improvement over the strongest adapted baseline) and F_1 (0.307), at roughly half the predicted set size of the neural baselines; the non-parametric EMPTRANS trails all learned models on set metrics. It also attains the best event NLL (2.89 vs. $\geq$2.96 for the other variants and $\approx$3.10 for the baselines) and is the first learned model in this comparison to reach the empirical transition table's single-type accuracy (0.243 vs. 0.248).

Next-Event Time, Type, and Volume

Table 2 reports target-free next-event timing, the single-next-event adaptation of Shi and Cartlidge (Shi and Cartlidge 2022) (primary label = a deterministic representative of the target set; event-type NLL from normalized channel intensities), and per-channel volume MAE (log1p space, true channels) from the joint volume head. VSM-RMTPP is the best-calibrated timer of all models: its target-free predictive median (0.098 s) essentially coincides with the true median inter-arrival (0.100 s) and its MAE (0.059) matches the best baseline. The HAWKES and S2P2 decoders are systematically *mis*-calibrated between events (predicted intensity $\approx 3\times$ the true rate; Sec.), while volume MAE is statistically tied across our variants and the best dedicated baseline head. One honest caveat: the inter-arrival distribution is nearly degenerate (the 0.1 s exchange grid concentrates much of the mass at exactly 0.100), so absolute timing differences are small.

[1]These are *adapted* backbones under a shared set-decoding protocol, not paper-faithful continuous-time likelihood implementations; comparisons are against the architectures' inductive biases, not the original systems.

Table 1: Event-set prediction on the test set (single seed; true mean set size 3.94). Threshold decode is shared by all models (τ calibrated on validation); EF_1 is the per-sample expected-F_1 decode. Best in **bold**.

Model	Jaccard↑	F_1 ↑	$\lvert\hat{S}\rvert$
Expected-F_1 decode			
VSM-HAWKES	0.183	0.289	7.4
VSM-RMTPP	0.181	0.286	7.8
VSM-S2P2	**0.202**	**0.307**	7.3
Shared global-threshold decode			
VSM-HAWKES	0.178	0.281	8.3
VSM-RMTPP	0.177	0.278	8.8
VSM-S2P2	**0.194**	**0.298**	8.9
Causal Transformer (SAHP-ad.)	0.185	0.286	9.3
Decayed cell (CT-LSTM-ad.)	0.187	0.288	9.9
PCT-LSTM (lite)	0.174	0.271	10.3
LSTM	0.182	0.285	11.1
EMPTRANS	0.165	0.265	6.0

Table 2: Target-free next-event time, Shi-style single-event metrics, and volume MAE (single seed). Our models read out the predictive median for MAE and predictive mean for RMSE. EMPTRANS NLL is omitted (non-comparable score family).

Model	MAE↓	RMSE↓	NLL↓	Acc.↑	Vol.MAE↓
VSM-HAWKES	0.108	0.199	2.96	0.202	0.720
VSM-RMTPP	**0.059**	0.194	2.97	0.221	0.729
VSM-S2P2	0.115	0.200	2.89	0.243	0.744
Causal Transformer (SAHP-ad.)	0.063	0.193	3.10	0.135	—
Decayed cell (CT-LSTM-ad.)	0.065	**0.192**	3.10	0.148	—
PCT-LSTM (lite)	0.064	**0.192**	3.12	0.153	—
LSTM	**0.059**	0.193	3.11	0.147	—
EMPTRANS	0.068	0.197	—	**0.248**	—

Stylized Facts of Simulated Order Flow

A generative model is only a *simulator* if it remains on-distribution when its own outputs are fed back as history. We evaluate this in closed loop: event sets, inter-arrival times, and per-channel volumes are sampled from the model (times by inversion of the compensator, sets from the conditional set distribution, volumes from the log-volume head) and appended to the conditioning history, for a fixed simulated *duration* (600 s) so that decoders with different event rates are compared over equal market time. Rollouts neither explode nor die out: simulated inter-arrival and set-size marginals remain at a constant distance from the real marginals across

Figure 3: Cont's eleven stylized facts, real vs. simulated order flow for VSM-RMTPP (single seed, 600 s closed-loop rollouts).

the full horizon.

Following the realism-evaluation practice of market simulators (Vyetrenko et al. 2020), we aggregate real and simulated streams into 1-second buckets and compute the eleven stylized facts of Cont (Cont 2001) on a signed order-flow proxy for returns (price-moving events: market orders and best-level in-spread insertions) and an activity series (events per bucket). Figure 3 shows the panel for VSM-RMTPP. The decoders separate sharply. VSM-RMTPP is the best all-round simulator: near-zero return autocorrelation (0.033 vs. 0.110 real), the closest tails (kurtosis 0.15-level), correct skew sign, a matching activity–volatility coupling (0.31 vs. 0.34 real), and the only non-degenerate volatility clustering signal (ACF$|r|$ 0.033 vs. 0.110 real). VSM-HAWKES under joint volume training simulates almost no volatility memory (0.007); an ablation without the volume head restores its clustering to 0.051 and its kurtosis to near-real (1.89 vs. 1.93), isolating a multi-task interference effect: the volume likelihood's gradients, flowing through the shared recurrent cell, dampen the *variability* of the self-exciting intensity while leaving prediction metrics unchanged. VSM-S2P2 is degenerate in closed loop (return autocorrelation 0.89: a near-deterministic drift), examined next.

Price-Level Realism

From events to prices. Simulated streams are converted to prices by a deterministic book-replay engine that inverts the event constructor's semantics: market orders consume the touch and walk the ladder; cancellations that empty the best level move the quote away; in-spread insertions improve it by their tick distance. Events within one timestamp are applied in matching-engine order (MO, IS, CO, LO). The engine assumes a dense one-tick ladder at the top of book; following the tick-size-regime analysis of Jain et al. (Jain et al. 2025), this is accurate for large-tick regimes and an approximation for small-tick assets, so we convert ladder units to prices via recorded per-level gaps and restrict price-level claims to distributional statements. The engine contains no stochastic or fitted components: unlike prior work that attaches order prices and volumes to simulated events from separately fitted power laws with hand-tuned aggression percentages (Shi and Cartlidge 2022), every stochastic quantity—type, side, book level, aggression, timing, volume—is produced by the model, and only deterministic mechanics are imposed.

Recorded ground truth. The real side of every comparison uses the *recorded* book state emitted by our event constructor (mid-price and full ten-level ladders per event), not a reconstruction. Simulated books are initialized from the recorded snapshot at the seed point. Model events that are inconsistent with the simulated book (cancellations of empty levels, insertions into a one-tick spread) are skipped and *counted*; we report this invalid-event rate as a measure of the model's structural coherence (VSM-HAWKES ??%, VSM-RMTPP ??%, VSM-S2P2 ??%).

Results. Against recorded reality (mean spread 0.3 bps), the simulated books separate sharply: VSM-RMTPP is the most faithful (0.1 bps, with zero impossible in-spread insertions), VSM-HAWKES is over-tight and over-calm (0.0 bps), and VSM-S2P2's book *diverges* to 10.2 bps mean (43.8 bps at p95) — $40\times$ reality. Thus the decoder that wins every one-step prediction column produces the least viable market: one-step predictive quality does *not* predict closed-loop viability, a compounding-error phenomenon familiar from model-based reinforcement learning that we propose as a first-class evaluation criterion — *closed-loop book stability* — for LOB generative models.

Discussion

Four findings shape how TFOW should be used. (1) *Set-valued modeling is the source of predictive advantage*: the state-space variant beats every adapted baseline on set metrics and matches the frequency table on single-type accuracy — multi-label structure, not next-type guessing, is where learnable signal lives. (2) *No decoder dominates; choose by task*: the state-space decoder for prediction and likelihood scoring, the constant-hazard recurrent decoder for timing and simulation — its rigid hazard is simultaneously why it cannot win the prediction table and why it is the only decoder whose closed-loop market remains faithful. (3) *Joint volume modeling is prediction-neutral but dampens self-exciting dynamics*: volume accuracy matches dedicated baseline heads with no cost to set/type/timing metrics, yet its gradients, flowing through the shared recurrent state, suppress the Hawkes decoder's closed-loop volatility clustering (restored by a volume-head-free ablation); we report this multi-task interference as a finding, and a stop-gradient volume readout as the remedy. (4) *Known limitations*: the feed's 0.1 s grid makes continuous hazards mis-

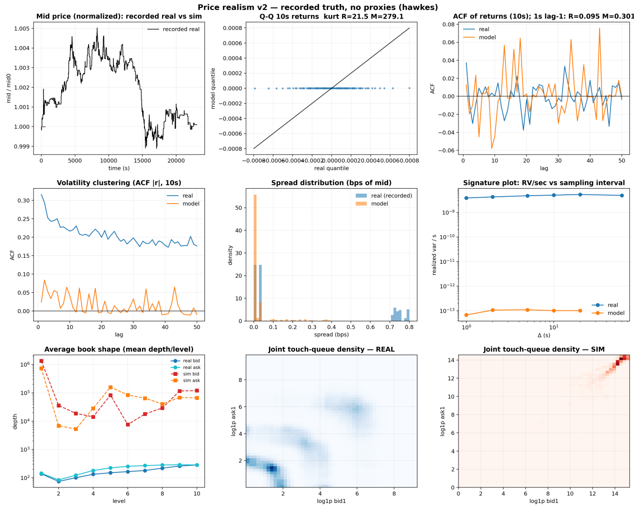

Figure 4: Price-level realism for VSM-HAWKES against recorded book state: mid trajectories, return Q-Q, autocorrelations, volatility clustering, spread, signature plot, book shape, and touch-queue density.

specified (the cause of both temporal PIT over-dispersion and the expressive decoders' between-event intensity inflation; a grid-aware discrete-time hazard is the principled fix), the conditionally independent set head caps within-tick co-occurrence structure, and the study covers a single venue, one week, and three symbols; scaling the corpus, a discrete-time temporal head, and a within-tick dependent set head are the subject of ongoing work. 1. M multiple scale, change the num and the scale nano second, check the scale

Related Work

Neural temporal point processes for LOB streams. Neural MTPPs (Mei and Eisner 2017; Du et al. 2016) model event streams with learned conditional intensities; Shi and Cartlidge (Shi and Cartlidge 2022) specialize them to LOB event streams with a market-state mechanism and parallel per-type intensity units, and evaluate both next-event prediction and simulation. We adopt their borrowed-backbone evaluation protocol and extend it in three directions: set-valued events with volumes (their model emits one event per timestamp; our data exhibits 3.94 simultaneous events on average), a fully target-free headline protocol (we additionally report the time-conditioned variant as a control), and a quantified realism battery (Cont's eleven facts plus price-level evaluation against recorded book state) in place of qualitative figure comparison. Their simulator attaches prices and volumes to model-generated (type, time) pairs by sampling separately fitted truncated power laws with hand-tuned market-order percentages; in our pipeline these quantities are model outputs. In the parametric Hawkes line, Jain et al. (Jain et al. 2024a) attach order sizes to compound Hawkes events from calibrated size distributions with time-of-day conditioning, validated by recovery of the concave market-impact function; our volume head replaces the calibrated size distribution with a *learned conditional* distribution, fitted jointly with set and timing likelihoods.

Set-valued point processes. Chang et al. (Chang, Boyd, and Smyth 2024b) introduce probabilistic models for sequences of sets in continuous time, which our volume-set MTPP builds on; we add continuous per-channel volumes and an explicit evaluation of the conditional-independence assumption of the Bernoulli set head, which our stylized-facts analysis identifies as the binding limitation for within-tick structure.

Market simulators and world models. Learned market simulators range from agent-based platforms (Byrd, Hybinette, and Balch 2020) to GAN-based LOB generators (Coletta et al. 2021) and, recently, order-level generative foundation models with controllable impact (MarS (Li et al. 2025)); see Jain et al. (Jain et al. 2024b) for a review of LOB simulation approaches. Relative to MarS, our model is not a foundation-scale transformer over order tokens but a set-valued MTPP with closed-form temporal statistics: the compensator supports time-rescaling goodness-of-fit (the basis of our 3S anomaly statistic) and probabilistic queries (hitting times, A-before-B) that token-level models do not expose. Realism evaluation against stylized facts follows (Vyetrenko et al. 2020); our addition is the closed-loop book-stability criterion and a recorded-truth protocol in which the real side of every comparison is free of reconstruction assumptions.

discussion and conclusion

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